

MUNDARE AGDM, AB, Canada

WMO: 713540

Lat: 53.5606N

Lon: 112.2961W

Elev: 2264

StdP: 13.53

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.3	-17.2	-29.1	1.2	-22.1	-23.4	1.6	-17.0	26.8	30.7	23.9	28.5	7.0	220	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	82.5	62.8	79.1	61.6	76.2	60.4	66.6	77.7	64.4	75.3	62.4	72.3	8.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.5	91.8	72.5	60.2	84.7	69.4	58.3	79.1	67.0	32.5	77.5	30.7	75.6	29.2	72.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	20.3	17.8	-29.3	88.2	4.4	3.1	-32.5	90.4	-35.0	92.2	-37.5	94.0	-40.7	96.2
			-29.5	71.0	4.3	2.2	-32.7	72.6	-35.2	73.8	-37.6	75.0	-40.8	76.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	12.9	12.5	23.3	38.4	51.1	58.2	62.5	60.0	51.6	39.0	23.1	12.2	(6)
	DBStd	21.51	15.29	14.34	12.81	9.40	7.91	5.37	5.06	5.78	8.10	8.67	12.66	14.25	(7)
	HDD50	5883	1149	1050	828	357	82	6	0	4	76	352	807	1172	(8)
	HDD65	10207	1614	1470	1293	797	433	211	110	172	407	806	1257	1637	(9)
	CDD50	1213	0	0	0	10	117	253	387	313	122	11	0	0	(10)
	CDD65	62	0	0	0	0	2	8	32	16	4	0	0	0	(11)
	CDH74	1133	0	0	0	2	88	145	444	304	148	2	0	0	(12)
	CDH80	221	0	0	0	0	11	16	103	55	36	0	0	0	(13)
	WSAvg	8.4	8.6	8.2	8.7	9.6	9.3	8.5	7.5	7.3	8.1	8.7	8.4	8.2	(14)
	PrecAvg	16.3	0.7	0.4	0.7	1.2	1.7	2.8	3.2	2.1	1.3	0.7	0.7	0.6	(15)
(16) Precipitation	PrecMax	22.6	2.7	0.9	1.5	2.6	4.6	5.0	5.5	4.1	3.2	1.7	3.0	1.5	(16)
	PrecMin	8.7	0.1	0.0	0.2	0.1	0.3	0.8	1.2	0.4	0.2	0.0	0.0	0.1	(17)
	PrecStd	2.9	0.6	0.2	0.3	0.6	0.9	1.1	1.1	0.7	0.8	0.4	0.6	0.4	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	43.6	44.3	52.7	71.7	81.9	82.3	87.8	85.9	85.2	71.2	54.6	44.2		(19)
	0.4% MCWB	36.9	37.0	42.4	50.5	56.8	61.9	66.4	63.3	60.2	52.9	44.2	37.5		(20)
	2% DB	40.1	39.6	45.9	65.0	76.2	78.2	82.7	80.7	78.2	62.1	46.8	39.1		(21)
	2% MCWB	34.7	34.0	38.5	47.7	54.3	59.6	65.9	62.9	58.0	48.8	39.2	33.9		(22)
	5% DB	37.2	35.9	42.1	59.1	71.6	74.6	78.8	77.4	72.4	56.6	41.9	35.5		(23)
	5% MCWB	33.0	32.0	36.1	44.5	52.4	58.7	64.0	62.2	56.1	45.2	36.2	31.4		(24)
	10% DB	33.7	32.4	38.6	54.1	67.3	71.0	75.4	73.8	66.3	52.3	38.1	31.7		(25)
	10% MCWB	30.6	29.3	33.9	42.0	50.4	56.7	62.5	60.9	53.4	43.3	34.0	28.8		(26)
	0.4% WB	37.5	37.6	43.5	51.4	58.2	65.4	70.9	68.4	61.9	53.8	44.5	37.7		(27)
	0.4% MCDB	42.2	43.3	51.0	70.0	76.8	77.3	81.6	78.1	77.8	67.6	53.0	43.2		(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	35.2	34.5	39.1	48.1	55.7	62.3	67.7	65.7	59.1	49.7	40.1	34.4		(29)
	2% MCDB	39.4	38.9	45.0	63.9	71.8	72.6	78.4	76.3	75.5	59.5	46.0	38.6		(30)
	5% WB	33.3	32.1	36.4	45.1	53.8	60.3	65.5	63.4	56.9	46.8	36.8	31.7		(31)
	5% MCDB	37.0	35.5	41.4	57.2	68.5	70.8	76.6	74.0	70.3	54.8	41.2	35.3		(32)
	10% WB	30.5	29.1	34.1	42.7	51.8	58.4	63.6	61.5	54.4	44.2	34.1	28.6		(33)
	10% MCDB	33.8	32.5	38.2	52.4	64.7	68.2	73.5	71.7	64.8	51.4	37.6	31.6		(34)
(36) Mean Daily Temperature Range	MDBR	16.3	17.1	16.5	19.5	22.7	20.7	21.3	22.3	22.2	18.8	15.2	15.1		(35)
	5% DB	17.0	20.8	19.6	29.7	29.2	26.6	27.5	29.1	32.8	26.8	19.2	19.0		(36)
	5% WB	13.1	16.3	13.6	15.2	12.8	11.9	13.5	14.3	14.9	15.3	13.0	15.1		(37)
	MCDBR	17.0	19.4	18.6	27.5	26.9	22.5	25.2	25.9	29.6	25.3	18.3	18.6		(38)
	MCWBR	13.6	15.7	13.1	14.8	12.3	11.4	13.5	13.9	14.3	14.9	13.2	15.3		(39)
(40) Clear-Sky Solar Irradiance	taub	0.254	0.259	0.306	0.330	0.345	0.360	0.358	0.369	0.308	0.276	0.251	0.241		(40)
	taud	2.303	2.350	2.316	2.364	2.377	2.358	2.394	2.333	2.488	2.549	2.443	2.280		(41)
	Ebn at Noon	234	274	279	286	286	282	280	270	278	266	239	214		(42)
	Edh at Noon	18	24	31	34	36	37	35	35	26	20	16	15		(43)
(44) All-Sky Solar Radiation	RadAvg	309	640	1129	1483	1824	1907	1954	1606	1113	640	306	228		(44)
	RadStd	35	49	87	139	133	126	97	113	116	78	35	22		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page